

Jalen Edusei

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Software developer shipping production, services-based web applications: REST services on PostgreSQL, React/TypeScript frontends, Docker/Kubernetes deployment, AI-paired development (Claude Code, Cursor) on Agile-style teams. HTML, CSS, JavaScript since 2022; Java since 2023.

EDUCATION

University of Georgia, Morehead Honors College, Athens, GA
Bachelor of Science in Computer Systems Engineering, cum laude

May 2026
GPA: 3.61 / 4.00

Relevant Coursework: Data Structures, Systems Programming, Software Development (Java), Discrete Mathematics, Embedded Systems I & II, Advanced Digital Design, Human-Computer Interaction

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, HTML, CSS, Python, SQL, Swift, C, C++, Elixir

Web & Services: React 19, Next.js, Node.js, Express, REST / JSON API design, WebSockets, JSON / YAML, Tailwind CSS

Data & Infrastructure: PostgreSQL (row-level security, transactional outbox), Supabase, SQLite, Docker, Kubernetes, GitHub Actions CI, Git / GitHub, Maven, Vercel

AI & Practices: AI-paired development (Claude Code, Cursor), Anthropic Claude API (tool use, streaming, structured JSON outputs), MCP servers, Agile team development, code review, automated testing (Vitest, ExUnit, XCTest)

EXPERIENCE

Freelance Software Engineer, *Self-Employed*, Atlanta, GA

May 2026 – Present

- Designed, built, and shipped kulenterprises.com for a Georgia freight carrier: 22 statically generated Next.js 16 / React 19 / TypeScript pages on a git-backed TinaCMS content model, a dependency-free site search, four rate-limited spam-protected forms through Resend, and WCAG AA-audited contrast.
- Extended the engagement with a broker credit-vetting portal on Supabase: a 12-question screening workflow, risk ratings with an admin-only do-not-load block, and compliance rules held in Postgres (row-level security, audit log, atomic SQL functions) behind invite-only auth with four staff roles.

Capital One, *Business Analyst Intern*, McLean, VA

Jun 2025 – Aug 2025

- Built the business case for a CreditWise Notifications Preferences Center projected to centralize communication management for 60M+ customer accounts; presented the recommendation to senior leadership.
- Analyzed email-campaign performance in SQL, Python, and Excel, creating a valuation framework that quantified engagement and retention impact and informed product-roadmap decisions.

UGA Small Satellite Research Lab, *Flight Software Developer (MEMESat-1)*, Athens, GA

Mar 2024 – Dec 2024

- Developed embedded C/C++ flight software on STM32 microcontrollers for the MEMESat-1 CubeSat, hardening it with unit and integration tests and reviewing teammates' code in the shared codebase.

PROJECTS

Relay: Order Management & Fulfillment | *Elixir, Phoenix, PostgreSQL, React, TypeScript, Docker, Kubernetes*

Jul 2026

- Built an event-driven order management system: a 13-route JSON API, a Kafka-style transactional outbox with offset-like sequencing, an 8-state machine enforced under row locks, a crash-safe worker with boot-time rehydration, and a live React dashboard over WebSockets.
- Prevented oversell with SELECT ...FOR UPDATE allocation across four facilities plus a CHECK constraint and Idempotency-Key replay; 52 failure-mode tests (ExUnit, Vitest) run in GitHub Actions CI; Docker and Kubernetes manifests with health probes.

Live Election Platform | *Next.js 14, JavaScript, Supabase (Postgres, Realtime, RLS)*

Mar 2026 – Jun 2026

- Shipped a realtime chapter election system (14 roles, 39 candidates): three-state election machine, SHA-256 device fingerprinting with a UNIQUE(role_id, device_hash) constraint and row-level security, and a 29-finding security audit verified by 42 HTTP-level tests.

AnimalDot Smart Bed (Senior Capstone) | *ESP32, C++, Express, PostgreSQL, MQTT, React Native*

Aug 2025 – May 2026

- Led all software (36 of 37 commits) on a six-person capstone team building a contactless pet vital-signs bed: ESP32 firmware with real-time DSP, a JWT-authenticated Express/PostgreSQL REST backend ingesting MQTT and broadcasting per-user over WebSockets, and React Native and SwiftUI apps.
- Designed a binary MQTT telemetry protocol with an embedded C++ encoder and independently tested TypeScript and Swift decoders, covering the monorepo with about 53 tests and two iOS CI pipelines.

Travel Itinerary Application | *Java 17, JavaFX, Maven, Gson, REST APIs*

Dec 2023

- Created a Java desktop application (Maven build) that assembles destination itineraries from the Google Places, REST Countries, and API Ninjas REST APIs, moving sequential HTTP calls onto a background thread so the interface stays responsive.

LEADERSHIP & HONORS

Vice President, NSBE UGA Chapter (100+ members) | Tau Beta Pi | Extraordinary Engineer Award | 2x Presidential Scholar | Dean's List